

Concept Selection, Prototyping & Testing

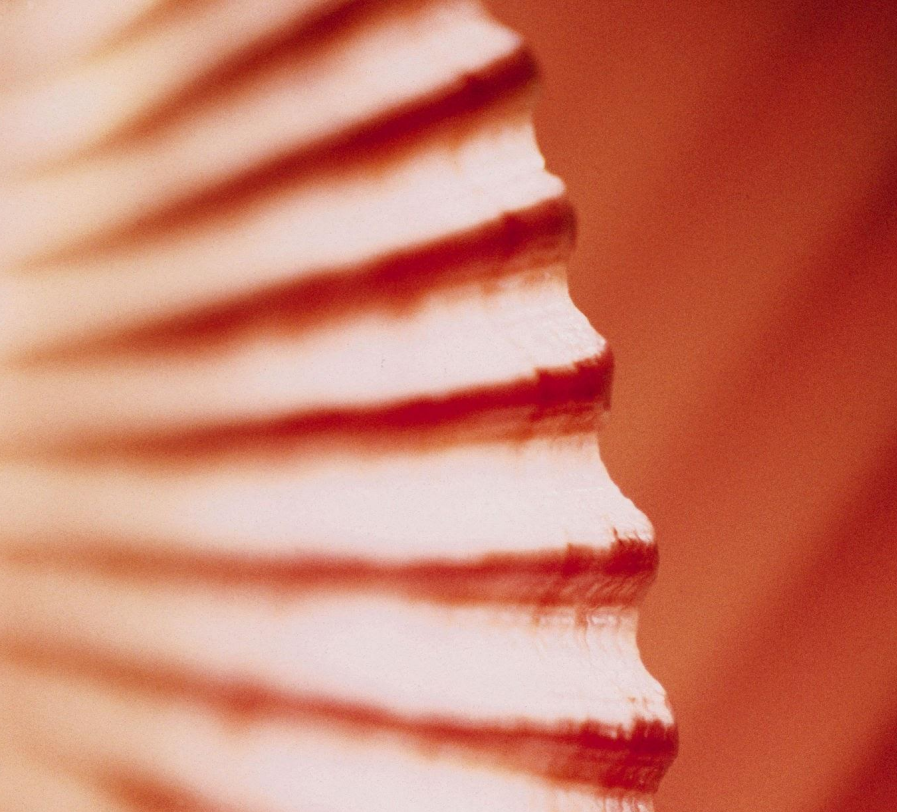
DESI 2701

SLIDES BY NZ GORHAM



Agenda

APRIL 2026



01 USER FEEDBACK SURVEY

02 PUGH CHART

03 LOW FIDELITY PROTOTYPE

04 PROTOTYPING TESTING

05 FINAL IDEA REFINEMENT

NOTE: There are times that I use “OHs”. OHs is the short term for Office Hours.

USER FEEDBACK SURVEY - SURVEY

APRIL 2026

BASIC INFORMATION

TARGET USERS: College Students

TARGET USER QUESTIONS:

- How many times do you go to Office Hours per week?
- Do you believe that going to Office Hours will help you reduce stress and success academically? Why or why not?

ADDITIONAL QUESTIONS:

- Which Idea is the most creative?
- Which Idea is the most feasible?

NUMBER OF RESPONSES: **14**

CONCEPT QUESTIONS

Question 1: On a scale of 1 to 10, how likely will you go to Office Hours if **IDEA #** existed?

Question 2: How many times will you go to Office Hours per week if **IDEA #** existed?

Question 3: How likely would you be using **IDEA #** over using AI tools (Chat GPT, Deek Seek, Chegg, Gemini, etc) to get help?

Question 4: Do you have any feedback for **IDEA #**? What improvements would you make on this idea?

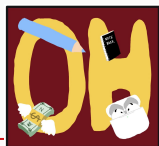
USER FEEDBACK SURVEY - RESULTS 1

APRIL 2026

IDEA 1: EXTRA OFFICE HOURS



IDEA 2: REWARD OFFICE HOURS



IDEA 3 - FLYERS IN THE AIR



IDEA 4 - UMN 1001



IDEA 5 - ELECTRO OFFICE HOURS



IDEA 6 - OFFICE HOURS FRATERNITY



Likely to go

Scale: 1: Not Likely - 10: Extremely Likely

Times you'll go to Office Hours per week

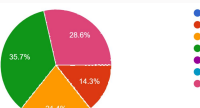
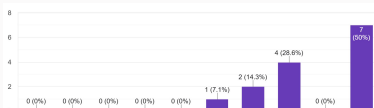
USE OFFICE HOURS OVER AI

Scale: 1: Not Likely - 5: Very Likely

FEEDBACK

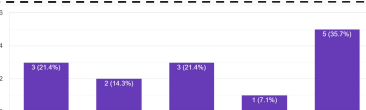
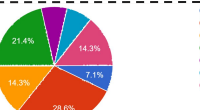
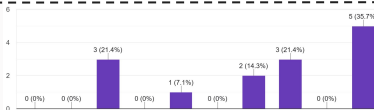
These are direct quotes

IDEA 1



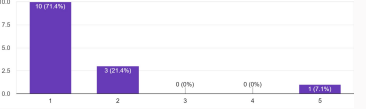
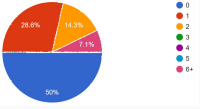
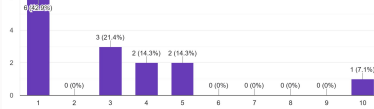
- I think it's a great idea to get students to office hours, but wouldn't really impact ai usage.
- I think it's a great idea I just don't think that teachers would like this
- I think this would be a great idea. However there would need to be some regulation on it. Maybe maxed out at like 5% with each session you attend being 0.5%.
- I like this idea a lot, it offers a good incentive.

IDEA 2



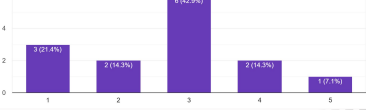
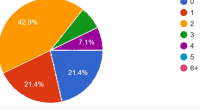
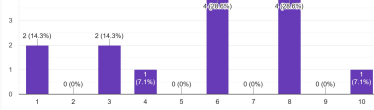
- Being able to get more than just Academic Items
- I don't think this would work because too many students would do it and it would be too expensive
- Just make sure the rewards are actually something the students want/need.
- If it were the case, this would be a great incentive to sit down and get work down and learn content!
- Cool idea but I usually put efficiency of my work and grade over stuff like rewards

IDEA 3



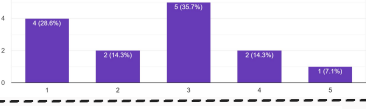
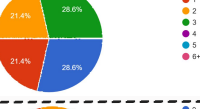
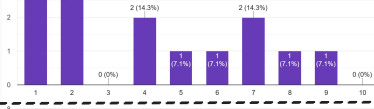
- hm it is good for awareness but just telling people to go won't help.
- I like the creativity of this, but since there isn't a reward there is low incentive to go.
- I think people would see it and ignore it because it seems like a fake add
- It's a fun idea but may not be realistic, and many people may not take it seriously

IDEA 4



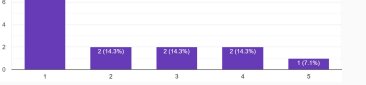
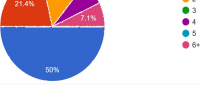
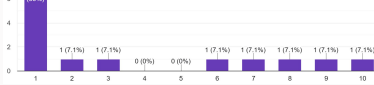
- Very nice idea, but maybe not making the class required would be better
- I like this idea. You get real advice from people older than you that have experienced more of college's expectations.
- I like this idea. I personally would use this class as a GPA booster opportunity, but this would definitely have a lot of benefit for students who are struggling academically, and the academic advisors can recommend this option for them.

IDEA 5



- Making sure the board is placed in high-traffic areas (like near entrances or elevators) so more students actually see it and use it.
- The biggest thing with office hours is not knowing when and where they are, and this makes it really accessible to know key information.
- I think it's a great idea to help spread awareness but wouldn't make me want to go

IDEA 6



- Im personally not a frat kind of person, but for the typical person this would be a great idea to bring people together and help encourage each other to work together to better their academic standings.
- making sure members are trained or approved so the help is reliable.
- There would be problems getting members for a house that would hit all the required classes plus I don't need a party for completing hw
- it is a great idea! But it wouldn't affect ME personally. I still might go and see, but often times I trust the teachers words way more than a tutor, and I don't really go to frats.

USER FEEDBACK SURVEY - RESULTS 2

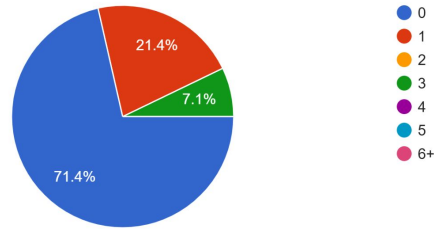
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TOP: BASIC QUESTIONS

BOTTOM: ADDITIONAL QUESTIONS

How many times do you go to Office Hours per week?

14 responses

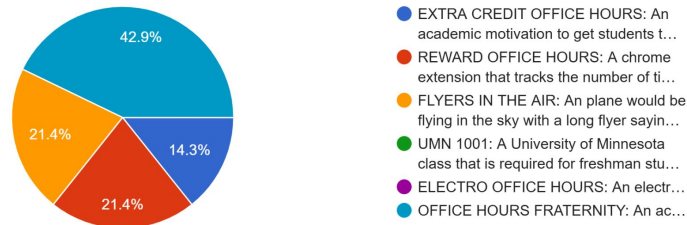


Do you believe that going to Office Hours will help you reduce stress and success academically? Why or why not?

- Yes. Even though I do not go I believe it is a great resource to have available for one on one time with your professors. I have utilized this time previously in classes where I feel I do not understand the content well, and it has helped tremendously.
- I do believe office hours are beneficial. You get the opportunity to ask specific questions pertaining to class content you personally are struggling with.
- Yes I think going to office hours can help reduce stress and improve academic success because it gives me a chance to ask questions directly and clear up confusion early.
- yes you can immediately ask questions to things you don't know, but you have to be honest with yourself and vulnerable to not know something.

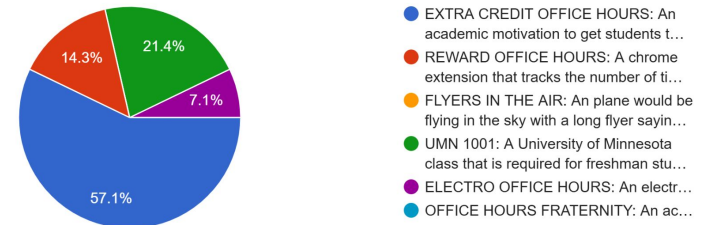
Which Idea is the most creative?

14 responses



Which Idea is the most feasible?

14 responses



PUGH CHART

APRIL 2026

FINAL DECISION

After doing the Pugh Chart, my final decision on the idea that I want to make is **IDEA 1: EXTRA CREDIT OFFICE HOURS** because it got the highest total value and I believe that this is the most realistic thing to make since it is fair to all students and doesn't fully violate the University of Minnesota's policy on passed everyone in the professor's class.

	EXTRA CREDIT OFFICE HOURS: An academic motivation to get students to go to Office Hours (OHs) to boost their grades up while getting the help they need to reduce their stress on hard assignments. In order to earn the extra credit, students need to show up and ask questions about assignments, exams, and topics related to the course.	REWARD OFFICE HOURS: A chrome extension that tracks the number of times you go to office hours and it will rack up points. Students can use those points to earn rewards like cookies, AirPods, iPads, Desktops, notebooks, or any other academic item. However, in order to earn points, students must work on assignments or over topics. They can't go to office hours and just socialize with the TA.	FLYERS IN THE AIR: An plane would be flying in the sky with a long flyer saying "GO TO OFFICE HOURS IF YOU WANT NO STRESS". It can also have Skydivers do this too. It is similar to the skydivers performing at the first UMN football game of the season	UMN 1001: A University of Minnesota class that is required for freshman students to take about it would have all students from all different departments and majors to learn about how to have a successful college experience and will teach students about Office Hours and why it is effective to your academic success and reduces stress. Students can take this class in any semester when they start their college journey.	ELECTRO OFFICE HOURS: An electric board that would display the schedule for Office Hours (OHs) for all classes where it is located along with updates on if office hours have been moved or cancelled. This is similar to Flight Information Display Systems. This should also hopefully encourage students going to OHs as well since there will be motivational text saying "PLEASE GO TO OFFICE HOURS! IT REDUCES STRESS AND WILL HELP YOU SUCCEED".	OFFICE HOURS FRATERNITY: An academic fraternity that will have members to host Office Hours from all class categories and will encourage students to get the help that they need. This will not be a social fraternity that throws a party 24/7 365, but when students do successful and consistently show growth and commitment, a non-alcoholic party can be thrown.
Novelty	1	2	0	0	1	1
Value	1	1	0	0	0	1
Feasibility	1	0	0	0	1	0
Theme	1	0	0	0	1	1
TOTAL	4	3	0	0	3	3
	1 - Standard	2 - Better	0 - Worst			

LOW FIDELITY PROTOTYPE - Sketch

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Description of the concept to build

What I am planning to build for my concept is an iPhone or iPad (most likely an iPhone) that will encourage students to go Office Hours

Anticipated resources and materials to make the model

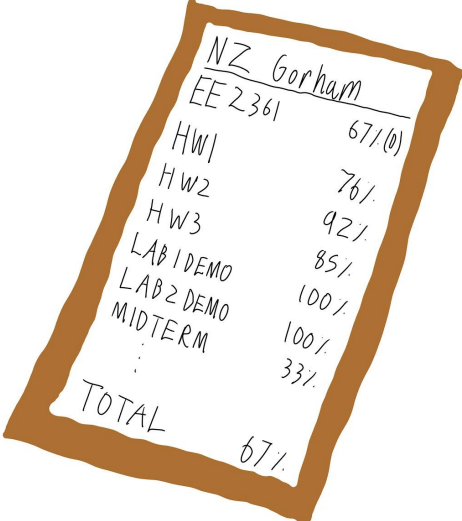
The resources and materials that I will use to create this is cardboard, glue, scissors, and paper and work in Anderson Labs

What you want to learn from testing the concept

I want to learn if my prototype will encourage students go to Office Hours so they can earn something good to pass the class and reduce stress academically

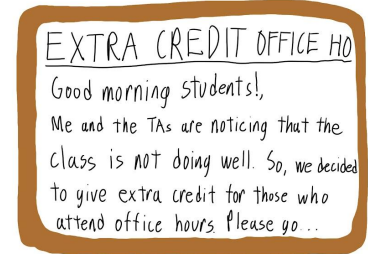
Who you plan to test with, where, and when

I am planning to test my prototype with my roommate Dublin on Wednesday at 12:45PM in the Bridges and Maryam at 2:45PM in McNeal Hall on Wednesday



A hand-drawn sketch of a tablet screen with a brown border. The screen displays a student's name 'NZ Gorham' and ID 'EE2361' at the top. Below is a list of assignments and their percentages: HW1 (67%), HW2 (76%), HW3 (92%), LAB1 DEMO (85%), LAB2 DEMO (100%), and MIDTERM (100%). At the bottom, it shows a 'TOTAL' of 67%.

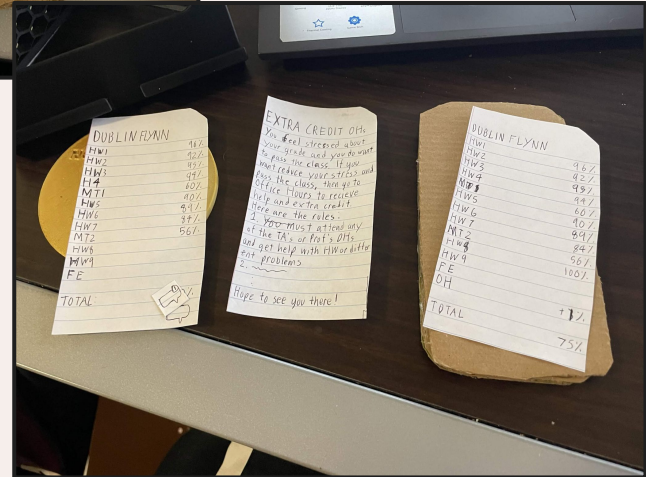
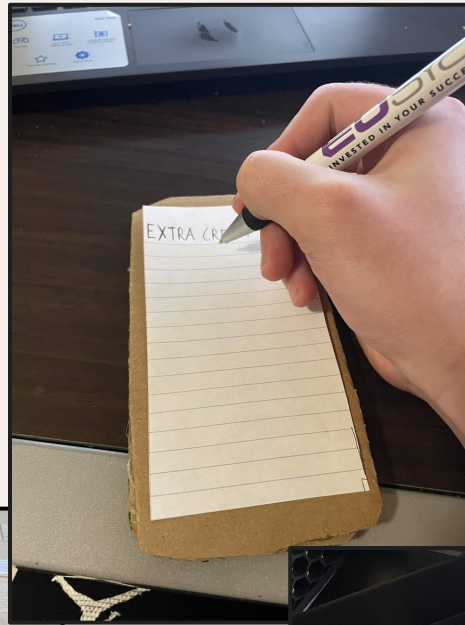
NZ Gorham EE2361	
HW1	67%
HW2	76%
HW3	92%
LAB1 DEMO	85%
LAB2 DEMO	100%
MIDTERM	100%
:	33%
TOTAL	67%



EXTRA CREDIT OFFICE
HOURS

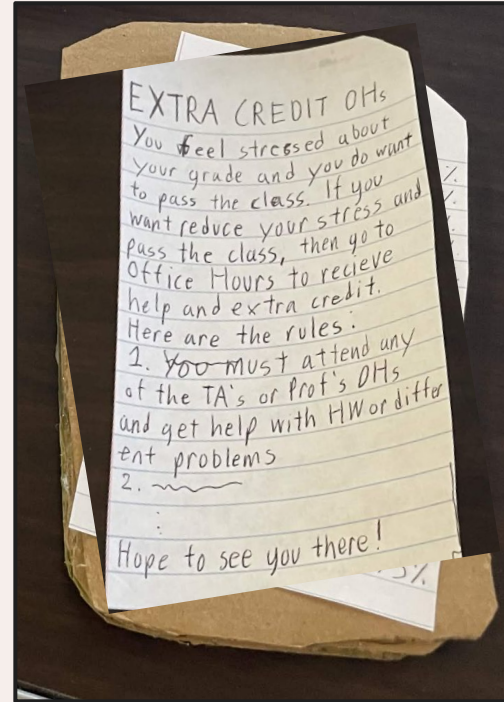
LOW FIDELITY PROTOTYPE - Work

APRIL 2026



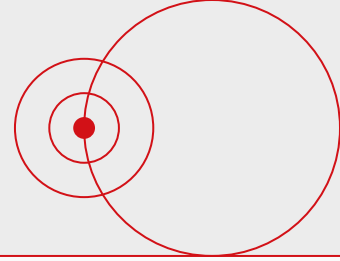
LOW FIDELITY PROTOTYPE - Prototype

My prototype ended up to be like an iPhone and what is supposed to happen during the testing part is that users would look at their grade and feel stressed that they are like at a 67-72% in the class and they want to pass the class without needing to retake it. They would get a message notifying them about Extra Credit Office Hours and hopefully it would encourage them to go to Office Hours so it can reduce their stress academically and receive extra credit in which is added to their final grade.



EXTRA CREDIT OFFICE
HOURS

PROTOTYPING TESTING



TESTER 1: DUBLIN FLYNN



What worked

- Being told that I get extra credit and gives my grade a boost

What did not work and could be improved

- Nothing

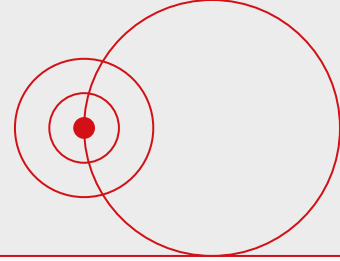
New ideas/suggestions that came up

- For the prototype, staple the papers together like a book

Questions that came up

- Will all OHs count as extra credit OHs and if not, when and why?
 - Yes, the more you go to OHs (it doesn't matter who it is), you the higher your extra credit will go

PROTOTYPING TESTING



TESTER 2: MARYAM JAMA



What worked

- The grades do stress me out and when I got the notification about the extra offices hours and the message was clear and it made me want to go to office hours

What did not work and could be improved

- Too many words the extra credit OHs page

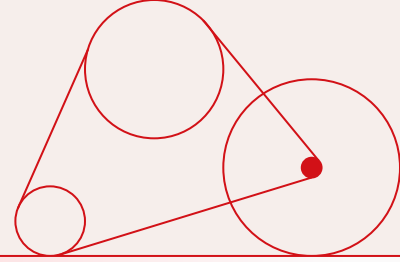
New ideas/suggestions that came up

- For the extra credit OHs page should have less words and more pictures/buttons that take you OHs pages

Questions that came up

- How would the extra credit be weighted in someone's grade?
 - It all depends on how many times the student go to OHs and them following the rules too

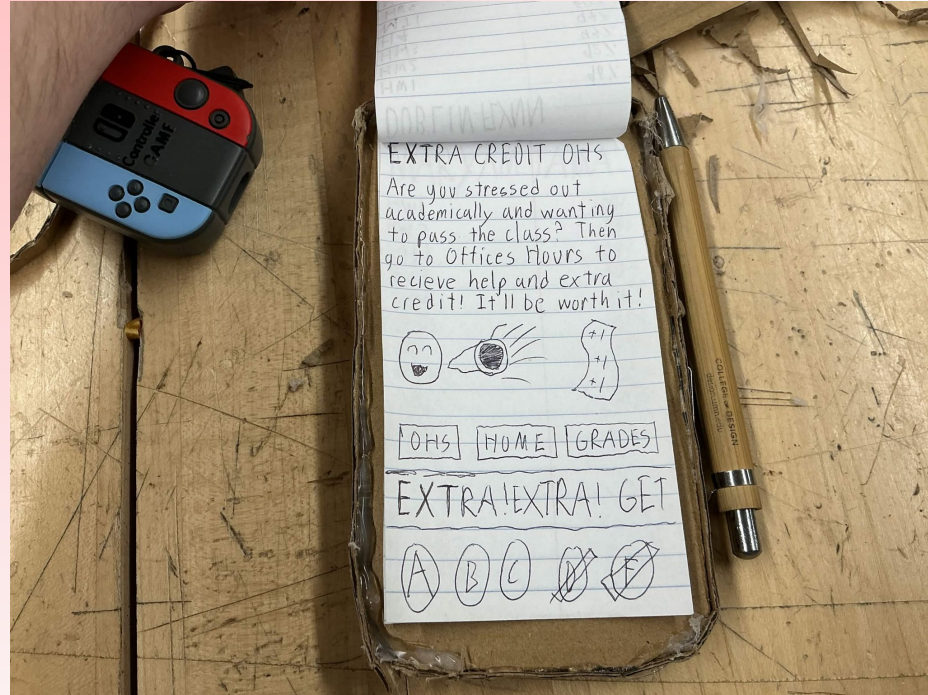
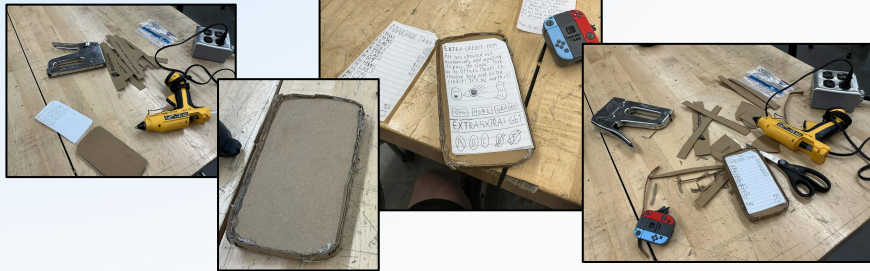
FINAL IDEA REFINEMENT

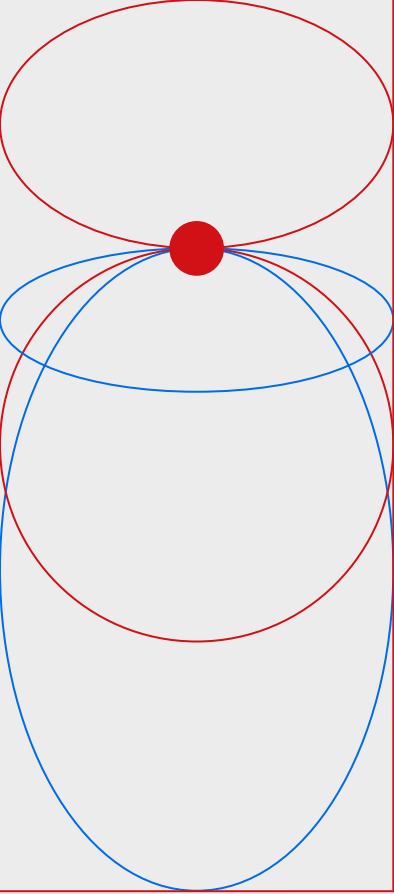


This is my updated and Final render for my concept/idea. The changes that I made were:

- Adding cardboard to the sides so it looks more cool - My Idea
- Less words and more pictures/buttons on the Extra Credit OHs page - Maryam's suggestion
- Stapling the papers together - Dublin's suggestion

Overall, I like how my final render turned out.





*Thank
you*